

Exact complex endpoint and C_0 norms, with strict interior phase loss

Candidate partial result for arXiv:2404.19540

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Status: candidate partial result; verifier verdict: likely valid. The result closes both L^p endpoints and the adjacent $C_0([0, 1])$ case for every complex order, and proves strict norm loss for every nonreal order in the interior range. It does not compute the square-space norm for general $1 < p < \infty$.

1 Source question

Alam, Chalendar, El Chami, Fricain, and Lefèvre study the analytic Riemann–Liouville semigroup in [1]. For $\xi \in \mathbb{C}$ with $\operatorname{Re} \xi > 0$, they write

$$(V_\xi f)(x) = \frac{1}{\Gamma(\xi)} \int_0^x (x-u)^{\xi-1} f(u) \, du, \quad 0 \leq x \leq 1. \quad (1)$$

Their $C_0([0, 1])$ denotes the continuous functions on $[0, 1]$ that vanish at 0.

Remark 1 on PDF page 14 gives

$$\|V_\xi\|_{\mathcal{B}(X)} \leq \frac{1}{\operatorname{Re}(\xi)|\Gamma(\xi)|}, \quad X = C_0([0, 1]) \text{ or } L^p(0, 1),$$

notes equality on L^1 when ξ is real, and then states that the exact L^p -norm is not known. The full source passage is reproduced below.

Thus (3.4) implies that

$$\|V_\xi(f)\|_q \leq K \frac{|\Gamma(\tau)|}{|\Gamma(\xi)|} \|f\|_p,$$

which shows that V_ξ is bounded from $L^p(0, 1)$ to $L^q(0, 1)$ and we have

$$\|V_\xi\|_{\mathcal{B}(L^p, L^q)} \leq K \frac{|\Gamma(\tau)|}{|\Gamma(\xi)|}.$$

(iv) Since $\xi \in \mathbb{C}_{\frac{1}{p}}$, then $\phi_\xi \in L^{p'}(0, 1)$. Thus $V_\xi(f) = f * \phi_\xi \in C_0([0, 1])$ with

$$\|V_\xi f\|_\infty \leq \|f\|_p \|\phi_\xi\|_{p'}.$$

Now, using (3.6), we get (3.5).

(v) Observe that $\phi_\xi \in C_0([0, 1])$ for $\operatorname{Re}(\xi) \geq 1$ with $\|\phi_\xi\|_\infty = \frac{1}{|\Gamma(\xi)|}$.

Then, argue as in (iv) with $p = 1$ and $p' = \infty$.

(vi) Use here $p = \infty$ and $p' = 1$, and argue as in (iv). $\square$

Remark 1. Let $X = C_0([0, 1])$ or $X = L^p([0, 1])$ for $1 \leq p \leq \infty$. As we have just seen, the operator V_ξ is bounded from X into itself and

$$(3.8) \quad \|V_\xi\|_{\mathcal{B}(X)} \leq \frac{1}{\operatorname{Re}(\xi)|\Gamma(\xi)|}.$$

Note that the estimate (3.8) for the case $X = C_0([0, 1])$ follows immediately from Proposition 3.1 and the fact that $C_0([0, 1])$ is contained in $L^\infty(0, 1)$. Observe also that for $X = L^p(0, 1)$, the estimate (3.8) can also be obtained easily by interpolation, or using the Minkowski inequality. Finally, it turns out that when $X = L^1(0, 1)$ and ξ is real, then (3.8) becomes an equality.

Note that in [1], J. Adell and E. Gallardo-Gutiérrez got some lower and upper bounds for the norm of the Riemann-Liouville operator V_t on $L^p(0, 1)$ when $t > 0$. These estimates enable them to get some asymptotic estimates of the norm. More precisely, they proved that, for every $1 \leq p \leq \infty$,

$$\lim_{t \rightarrow \infty} \Gamma(t+1) \|V_t\|_{\mathcal{B}(L^p)} = c_{p,q}$$

where

$$c_{p,q} = \begin{cases} p^{-1/p} q^{-1/q} & \text{if } 1 < p < \infty \\ 1 & \text{if } p = 1 \text{ or } p = \infty. \end{cases}$$

However, it should be noted that the exact computation of $\|V_\xi\|_{\mathcal{B}(L^p)}$ is not known.

Figure 1: Proposition 3.1(iv)–(vi) and Remark 1 of [1], PDF page 14. The crop retains the complete exact-norm statement.

2 Proof intuition

The kernel modulus depends only on $x - u$. At the L^∞ endpoint, one may align the phase of the single row at $x = 1$, so the row's L^1 mass is the exact norm. At the L^1 endpoint, normalized mass concentrating at $u = 0$ makes the output converge to the first kernel column; the column has the same L^1 mass. These are genuinely endpoint mechanisms and do not produce an exact interior L^p -norm.

For the C_0 restriction, the phase-aligning L^∞ function can be cut off continuously near both endpoints. The lost kernel mass tends to zero. More generally, the exact $L^p \rightarrow C_0$ norm is simply the $L^{p'}$ -norm of the row at $x = 1$, because Hölder gives this as an upper bound and evaluation at 1 supplies the matching lower bound.

In the interior range, compactness guarantees an extremizer. If a nonreal order attained the natural domination bound by its real-order counterpart, equality in the triangle inequality would force the oscillatory phases $(x - u)^{i\eta}$ to align simultaneously for almost every pair (x, u) . Two distinct input points make that impossible as x varies. This gives a strict comparison, though not the still-unknown exact interior constant.

3 Exact endpoint theorem

Put $\tau = \operatorname{Re} \xi$ and $\eta = \operatorname{Im} \xi$.

Theorem 1. *Let $\xi = \tau + i\eta \in \mathbb{C}$ with $\tau > 0$, and let V_ξ be given by (1).*

(i) *The exact square-space endpoint and C_0 norms are*

$$\begin{aligned} \|V_\xi\|_{L^1 \rightarrow L^1} &= \|V_\xi\|_{L^\infty \rightarrow L^\infty} = \|V_\xi\|_{L^\infty \rightarrow C_0} = \|V_\xi\|_{C_0 \rightarrow C_0} \\ &= \frac{1}{\tau |\Gamma(\xi)|}. \end{aligned}$$

(ii) *Let $1 < p \leq \infty$, let p' be conjugate to p , and assume $\tau > 1/p$. Then*

$$\|V_\xi\|_{L^p \rightarrow C_0} = \frac{1}{|\Gamma(\xi)|((\tau - 1)p' + 1)^{1/p'}}. \quad (2)$$

For $p = \infty$, the convention is $p' = 1$, so (2) reduces to $1/(\tau |\Gamma(\xi)|)$.

(iii) *The map $V_\xi : L^1(0, 1) \rightarrow C_0([0, 1])$ is bounded if and only if $\tau \geq 1$. In that case*

$$\|V_\xi\|_{L^1 \rightarrow C_0} = \frac{1}{|\Gamma(\xi)|}. \quad (3)$$

Proof. For $0 < x \leq 1$, write the kernel row as

$$k_x(u) = \frac{\mathbf{1}_{(0,x)}(u)(x - u)^{\xi-1}}{\Gamma(\xi)}.$$

Its modulus is

$$|k_x(u)| = \frac{\mathbf{1}_{(0,x)}(u)(x-u)^{\tau-1}}{|\Gamma(\xi)|}. \quad (4)$$

The $L^p \rightarrow C_0$ row norm. Assume first that $1 < p \leq \infty$ and $\tau > 1/p$. Hölder's inequality and (4) give

$$|V_\xi f(x)| \leq \|k_x\|_{p'} \|f\|_p, \quad (5)$$

$$\|k_x\|_{p'} = \frac{x^{\tau-1/p}}{|\Gamma(\xi)|((\tau-1)p' + 1)^{1/p'}}. \quad (6)$$

The exponent $\tau - 1/p$ is positive, so the right side is largest at $x = 1$. This proves the upper bound in (2). Conversely, evaluation at $x = 1$ shows that the operator norm is at least the norm of the functional $f \mapsto \int_0^1 k_1(u)f(u) du$, namely $\|k_1\|_{p'}$. Thus equality holds.

For completeness, the image really lies in the source's C_0 . Extend f and $s \mapsto \mathbf{1}_{(0,1)}(s)s^{\xi-1}/\Gamma(\xi)$ by zero to $\mathbb{R}$. For finite p , translation continuity of f in $L^p(\mathbb{R})$, followed by Hölder, makes the convolution continuous. For $p = \infty$, use translation continuity of the kernel in $L^1(\mathbb{R})$. Its value at 0 is zero by the Volterra support.

The same row argument at $p = 1$ uses the L^∞ -norm of k_x . It is uniformly finite exactly when $\tau \geq 1$, and then

$$\sup_{0 < x \leq 1} \|k_x\|_\infty = \frac{1}{|\Gamma(\xi)|}.$$

This proves (3). If $\tau < 1$, the functional associated to k_1 is unbounded on L^1 ; inputs supported on intervals approaching $u = 1$ give an explicit unbounded sequence. When $\tau \geq 1$, continuity of the output follows by translating the zero extension of f in $L^1(\mathbb{R})$ against the bounded kernel. This also covers the boundary $\tau = 1$, even when $\eta \neq 0$.

The square L^1 -norm. Tonelli's theorem yields, for every $f \in L^1(0, 1)$,

$$\begin{aligned} \|V_\xi f\|_1 &\leq \frac{1}{|\Gamma(\xi)|} \int_0^1 |f(u)| \int_u^1 (x-u)^{\tau-1} dx du \\ &= \frac{1}{\tau|\Gamma(\xi)|} \int_0^1 (1-u)^\tau |f(u)| du \\ &\leq \frac{1}{\tau|\Gamma(\xi)|} \|f\|_1. \end{aligned}$$

To obtain the reverse inequality, set

$$h(s) = \frac{\mathbf{1}_{(0,1)}(s)s^{\xi-1}}{\Gamma(\xi)} \quad (s \in \mathbb{R}), \quad f_\varepsilon = \varepsilon^{-1} \mathbf{1}_{(0,\varepsilon)}.$$

Then $h \in L^1(\mathbb{R})$, $\|f_\varepsilon\|_1 = 1$, and on $(0, 1)$

$$V_\xi f_\varepsilon = \frac{1}{\varepsilon} \int_0^\varepsilon h(\cdot - u) du.$$

Translation continuity in $L^1(\mathbb{R})$ therefore gives $V_\xi f_\varepsilon \rightarrow h$ in $L^1(0, 1)$. Consequently

$$\|V_\xi\|_{L^1 \rightarrow L^1} \geq \|h\|_{L^1(0,1)} = \frac{1}{\tau|\Gamma(\xi)|},$$

which proves the exact L^1 -norm.

The L^∞ and C_0 square norms. Taking $p = \infty$ in the already proved row formula gives both

$$\|V_\xi\|_{L^\infty \rightarrow C_0} = \|V_\xi\|_{L^\infty \rightarrow L^\infty} = \frac{1}{\tau|\Gamma(\xi)|}.$$

Restriction of the domain gives the same number as an upper bound on $C_0 \rightarrow C_0$. For the lower bound, choose continuous functions $0 \leq \chi_n \leq 1$, supported in $(0, 1)$, with $\chi_n(u) \rightarrow 1$ for every $u \in (0, 1)$, and define

$$f_n(u) = \chi_n(u) \frac{\Gamma(\xi)}{|\Gamma(\xi)|} (1-u)^{-i\eta} \quad (0 \leq u < 1), \quad f_n(1) = 0.$$

Because χ_n vanishes near both endpoints, $f_n \in C_0([0, 1])$ and $\|f_n\|_\infty \leq 1$. The phases align at $x = 1$, so

$$V_\xi f_n(1) = \frac{1}{|\Gamma(\xi)|} \int_0^1 (1-u)^{\tau-1} \chi_n(u) du \longrightarrow \frac{1}{\tau|\Gamma(\xi)|}$$

by dominated convergence. This is the matching lower bound on $C_0 \rightarrow C_0$ and completes the proof. $\square$

4 Strict interior phase loss

The full exact interior norm remains out of reach, but complex orders admit a strict comparison with the corresponding positive real-order operator.

Theorem 2 (Strict complex-to-real comparison). *Let $1 < p < \infty$, $\tau > 0$, and $\eta \in \mathbb{R} \setminus \{0\}$. Then*

$$\|V_{\tau+i\eta}\|_{L^p \rightarrow L^p} < \frac{\Gamma(\tau)}{|\Gamma(\tau+i\eta)|} \|V_\tau\|_{L^p \rightarrow L^p}. \quad (7)$$

Moreover, for every $\xi \in \mathbb{C}$ with $\operatorname{Re} \xi = \tau > 0$,

$$\|V_\xi\|_{L^p \rightarrow L^p} < \frac{1}{\tau|\Gamma(\xi)|}. \quad (8)$$

Proof. We first record compactness. Extend

$$h_\xi(s) = \frac{\mathbf{1}_{(0,1)}(s)s^{\xi-1}}{\Gamma(\xi)}$$

by zero to $\mathbb{R}$. Approximate h_ξ in $L^1(\mathbb{R})$ by continuous kernels supported away from 0. The corresponding integral operators on $[0, 1]$ have continuous kernels and are norm limits of finite-rank operators, hence are compact. Young's inequality transfers the L^1 -kernel approximation to operator norm. Thus V_ξ is compact on $L^p(0, 1)$. Since L^p is reflexive, V_ξ attains its norm.

Pointwise domination gives

$$|V_{\tau+i\eta}f(x)| \leq \frac{\Gamma(\tau)}{|\Gamma(\tau+i\eta)|} V_\tau(|f|)(x). \quad (9)$$

Suppose equality held in (7), and let $f \in L^p$, $\|f\|_p = 1$, attain the norm of $V_{\tau+i\eta}$. Both inequalities in

$$\|V_{\tau+i\eta}f\|_p \leq \frac{\Gamma(\tau)}{|\Gamma(\tau+i\eta)|} \|V_\tau|f|\|_p \leq \frac{\Gamma(\tau)}{|\Gamma(\tau+i\eta)|} \|V_\tau\|_{p \rightarrow p}$$

would then be equalities. In particular, equality holds in (9) for almost every x .

Let $E = \{u : f(u) \neq 0\}$, which has positive measure, and write $\zeta(u) = f(u)/|f(u)|$ on E . Equality in the triangle inequality for the defining integral implies that, for almost every x , there is a unimodular number $\omega(x)$ such that

$$e^{i\eta \log(x-u)} \zeta(u) = \omega(x) \quad (10)$$

for almost every $u \in E \cap (0, x)$. By Fubini, for almost every pair $(u, v) \in E^2$ and almost every $x > \max\{u, v\}$,

$$e^{i\eta[\log(x-u)-\log(x-v)]} = \frac{\zeta(v)}{\zeta(u)}.$$

Choose such a pair with $u \neq v$. The left side is continuous in x , so equality almost everywhere on $(\max\{u, v\}, 1)$ makes it constant there. Its logarithmic derivative is

$$i\eta \left(\frac{1}{x-u} - \frac{1}{x-v} \right),$$

which is nonzero. This contradiction proves (7).

It remains to prove (8) for real order. The convolution bound gives $\|V_\tau\|_{p \rightarrow p} \leq 1/\Gamma(\tau+1)$. If equality held, compactness would give a norming $g \geq 0$ with $\|g\|_p = 1$. Set

$$(S_s g)(x) = \mathbf{1}_{(s,1)}(x) g(x-s).$$

Minkowski's inequality would give

$$\begin{aligned} \|V_\tau g\|_p &\leq \frac{1}{\Gamma(\tau)} \int_0^1 s^{\tau-1} \|S_s g\|_p ds, \\ \|S_s g\|_p^p &= \int_0^{1-s} |g(u)|^p du. \end{aligned}$$

For s sufficiently close to 1, the latter norm is strictly less than one. Since the weight is positive, the integral is strictly less than $1/\Gamma(\tau+1)$, a contradiction. Hence the real-order Young bound is strict. For nonreal order, combine this bound with (7); (8) follows in all cases. $\square$

5 What remains open

Theorem 1 settles the exact square norm for $p = 1, \infty$ and the complete C_0 analogue. It also turns the source's $L^p \rightarrow C_0$ estimates into equalities. The general exact value of $\|V_\xi\|_{L^p \rightarrow L^p}$ for $1 < p < \infty$ remains open.

Theorem 2 shows that the imaginary part cannot be removed by a phase choice: nonreal orders incur a strict loss relative to the real-order domination constant. It also proves that the elementary kernel bound is never sharp in the interior. But the positive real-order norm $\|V_\tau\|_{L^p \rightarrow L^p}$ is itself unknown for general τ , so the strict comparison does not yield an exact numerical value.

The endpoint constant cannot be extended to interior p merely by interpolation: for the classical Volterra operator V_1 on $L^2(0, 1)$, the exact norm is $2/\pi$, whereas $1/(\operatorname{Re}(1)|\Gamma(1)|) = 1$.

6 Verification and novelty status

The proofs were checked line by line against the main likely failure points: the complex phase in the L^1 column limit, continuity into C_0 at $\tau = 1$, the endpoint cutoffs in the C_0 lower bound, compactness and norm attainment in the interior, and the Fubini phase-rigidity step. All checks pass. The companion scripts numerically evaluate the explicit lower bound families and a discretized strict-comparison example; they are only sanity checks, not part of the proof. Run them with

```
conda run --no-capture-output -n sandbox python \
  code/check_endpoint_norms.py
conda run --no-capture-output -n sandbox python \
  code/check_strict_phase_loss.py
```

A bounded search covered the run indexes; the local parsed source series arXiv:2108.02291, 2112.02669, 2408.10180, and 2410.00830; and arXiv/DOI web queries for complex-order endpoint norms. Carvalho-Neto and Fehlberg Júnior record the real-order L^1 and L^∞ endpoint formula in [2]. No complex-order endpoint or C_0 formula was found. The search also included strict complex-to-real norm comparison and imaginary-order phase-loss phrases, with no exact match found. Novelty confidence remains moderate because the proof is elementary and the claim may be folklore.

Human-review recommendation. Verify the endpoint norm-saturating arguments and the Fubini phase-rigidity step, then perform a specialist literature check before making any external originality claim.

References

- [1] I. Alam, I. Chalendar, F. El Chami, E. Fricain, and P. Lefèvre, *Eventual Ideal Properties of the Riemann–Liouville Analytic Semigroup*, arXiv:2404.19540v2, 2026; Remark 1, p. 14.
- [2] P. M. Carvalho-Neto and R. Fehlberg Júnior, *The Riemann–Liouville fractional integral in Bochner–Lebesgue spaces I*, arXiv:2108.02291v2, 2021; Theorem 4.1.